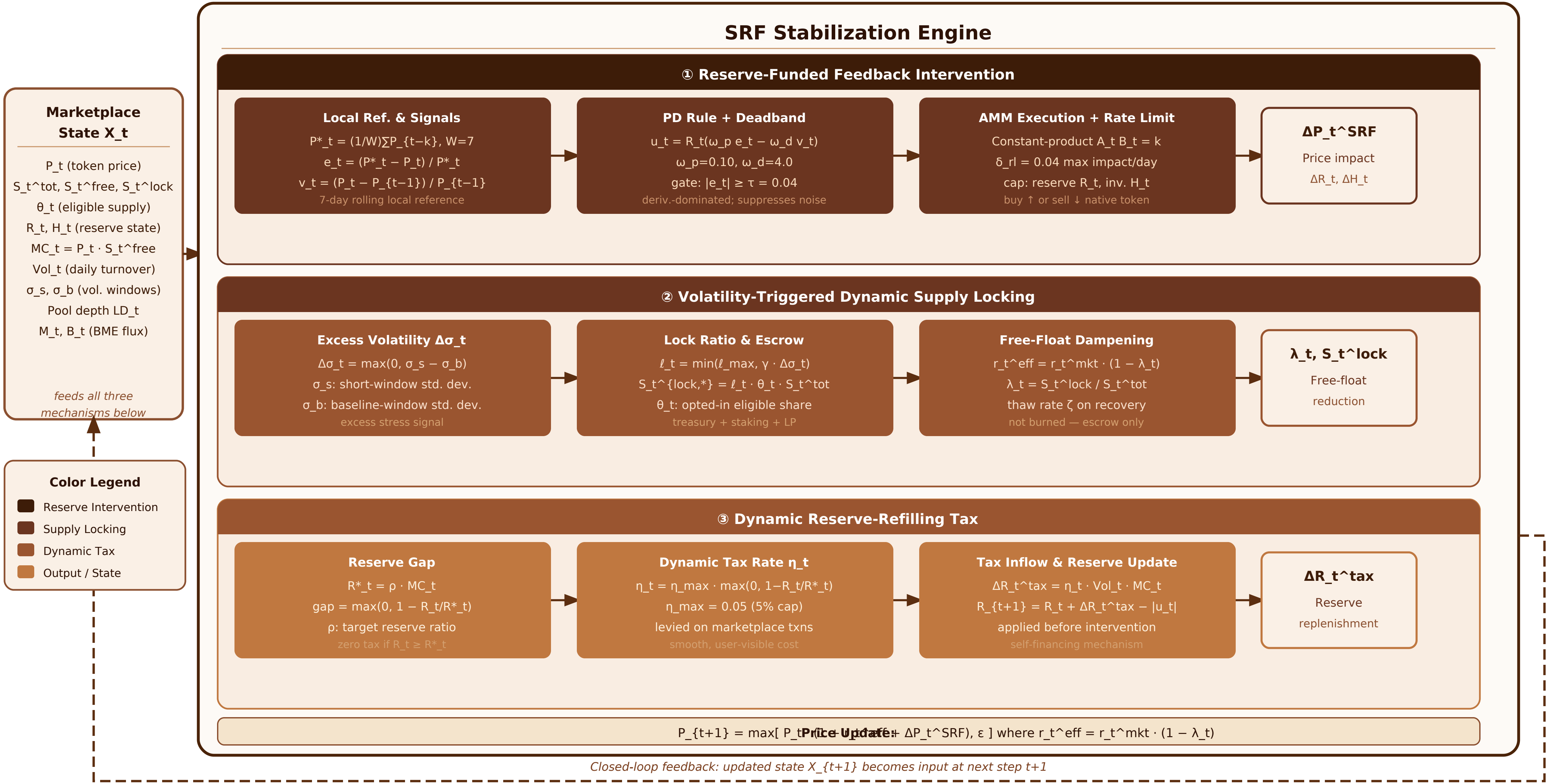
SRF Stabilization Engine
① Reserve-Funded Feedback Intervention
Local Ref. & Signals
P*_t = (1/W)∑P_{t−k}, W=7
e_t = (P*_t − P_t) / P*_t
v_t = (P_t − P_{t−1}) / P_{t−1}
7-day rolling local reference
PD Rule + Deadband
u_t = R_t(ω_p e_t − ω_d v_t)
ω_p=0.10, ω_d=4.0
gate: |e_t| ≥ τ = 0.04
deriv.-dominated; suppresses noise
AMM Execution + Rate Limit
Constant-product A_t B_t = k
δ_rl = 0.04 max impact/day
cap: reserve R_t, inv. H_t
buy ↑ or sell ↓ native token
ΔP_t^SRF
Price impact
ΔR_t, ΔH_t
② Volatility-Triggered Dynamic Supply Locking
Excess Volatility Δσ_t
Δσ_t = max(0, σ_s − σ_b)
σ_s: short-window std. dev.
σ_b: baseline-window std. dev.
excess stress signal
Lock Ratio & Escrow
ℓ_t = min(ℓ_max, γ · Δσ_t)
S_t^{lock,*} = ℓ_t · θ_t · S_t^tot
θ_t: opted-in eligible share
treasury + staking + LP
Free-Float Dampening
r_t^eff = r_t^mkt · (1 − λ_t)
λ_t = S_t^lock / S_t^tot
thaw rate ζ on recovery
not burned — escrow only
λ_t, S_t^lock
Free-float
reduction
③ Dynamic Reserve-Refilling Tax
Reserve Gap
R*_t = ρ · MC_t
gap = max(0, 1 − R_t/R*_t)
ρ: target reserve ratio
zero tax if R_t ≥ R*_t
Dynamic Tax Rate η_t
η_t = η_max · max(0, 1−R_t/R*_t)
η_max = 0.05 (5% cap)
levied on marketplace txns
smooth, user-visible cost
Tax Inflow & Reserve Update
ΔR_t^tax = η_t · Vol_t · MC_t
R_{t+1} = R_t + ΔR_t^tax − |u_t|
applied before intervention
self-financing mechanism
ΔR_t^tax
Reserve
replenishment
Price Update:
P_{t+1} = max[P_t(1 + r_t^eff + ΔP_t^SRF), ε] where r_t^eff = r_t^mkt · (1 − λ_t)
Marketplace
State X_t
P_t (token price)
S_t^tot, S_t^free, S_t^lock
θ_t (eligible supply)
R_t, H_t (reserve state)
MC_t = P_t · S_t^free
Vol_t (daily turnover)
σ_s, σ_b (vol. windows)
Pool depth LD_t
M_t, B_t (BME flux)
feeds all three
mechanisms below
Color Legend
Reserve Intervention
Supply Locking
Dynamic Tax
Output / State
Closed-loop feedback: updated state X_{t+1} becomes input at next step t+1